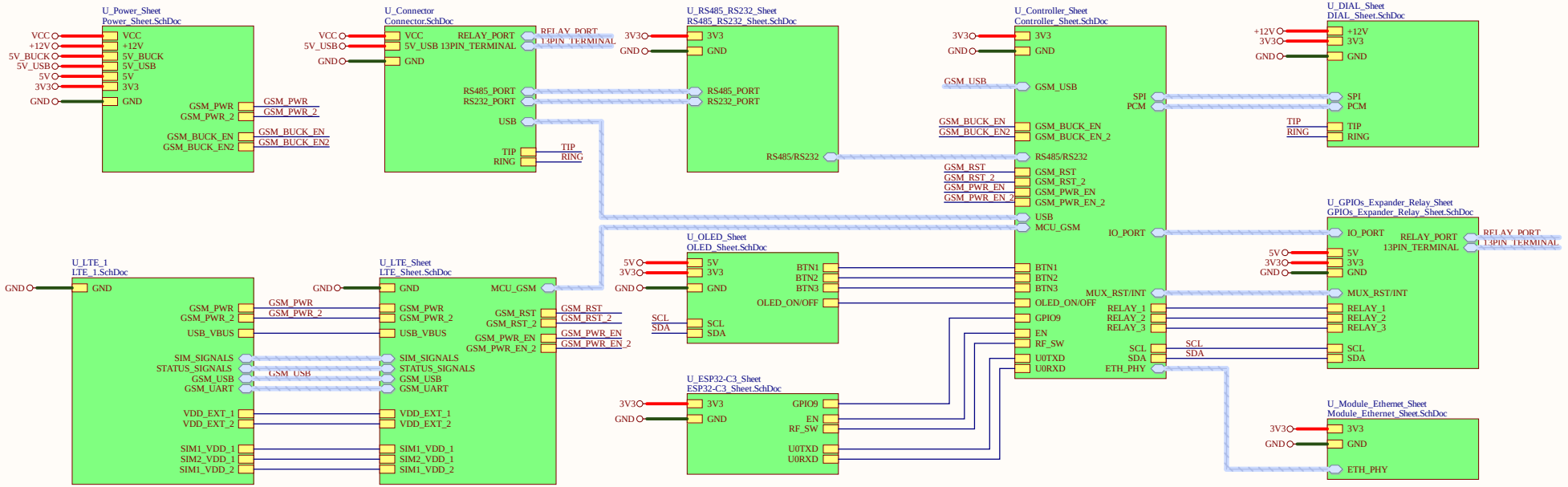
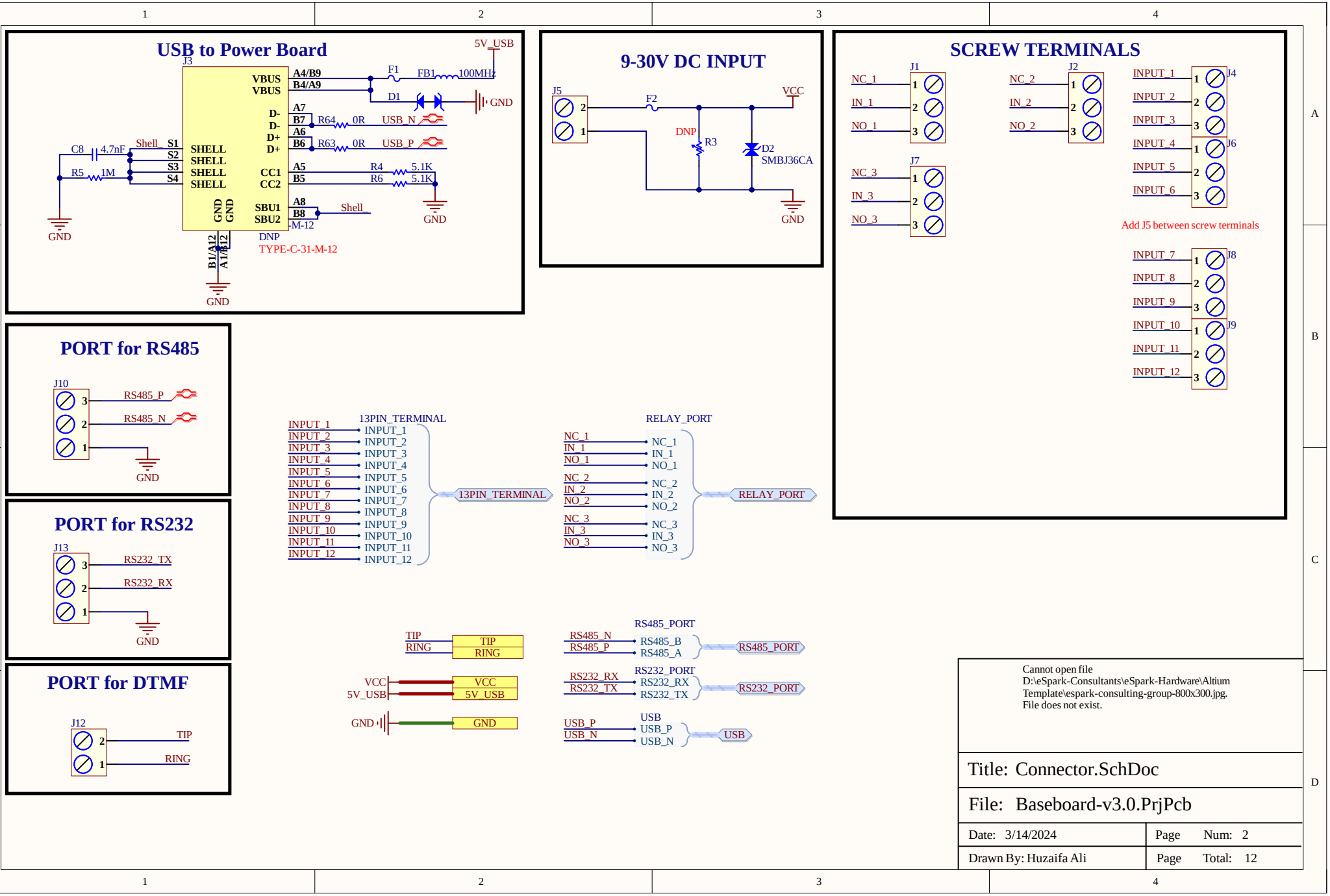


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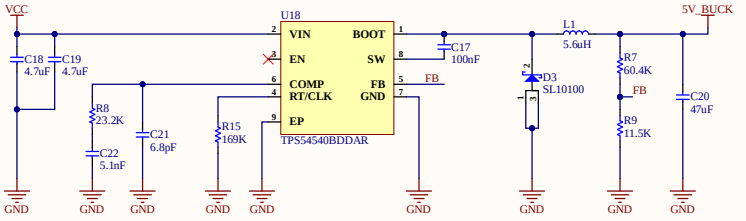
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File: Baseboard-v3.0.PrjPcb	
Date: 3/19/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 12



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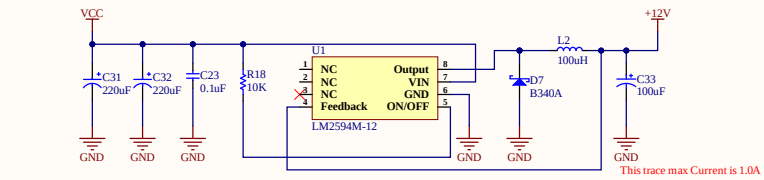
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Date: 3/14/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 12

5V BUCK REGULATOR



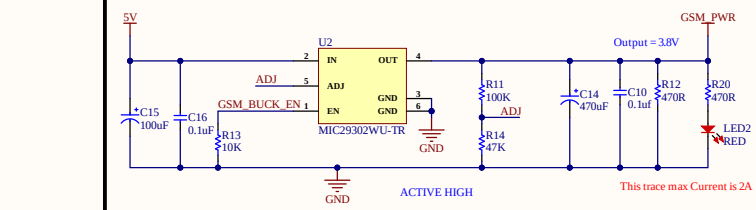
This trace max Current is 5A

12V BUCK REGULATOR



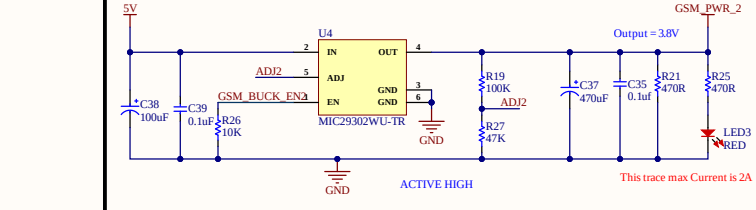
This trace max Current is 1.0A

BUCK FOR GSM POWER



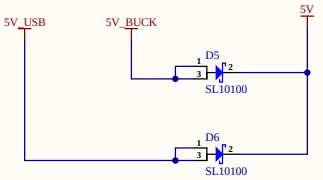
This trace max Current is 2A

BUCK 2 FOR GSM POWER 2



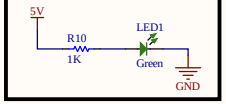
This trace max Current is 2A

ORING CIRCUIT

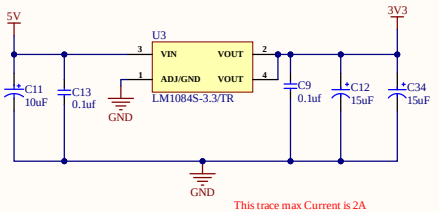


This trace max Current is 5A

POWER LED

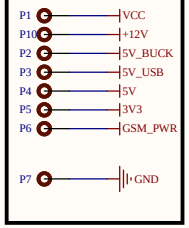


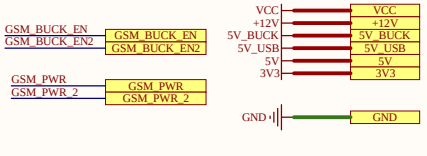
LDO 3.3V



This trace max Current is 2A

TEST POINTS

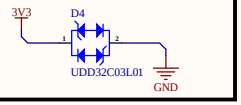




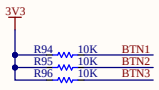
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File: Baseboard-v3.0.PrjPcb	
Date: 3/19/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 12

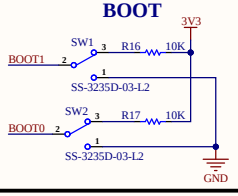
ESD PROTECTION



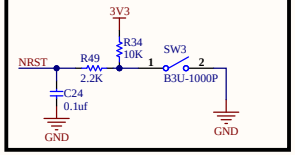
PULL-UP



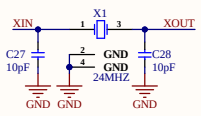
BOOT



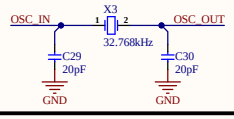
RESET

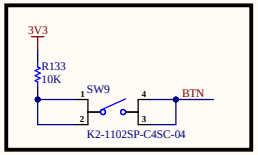


24MHz CRYSTAL

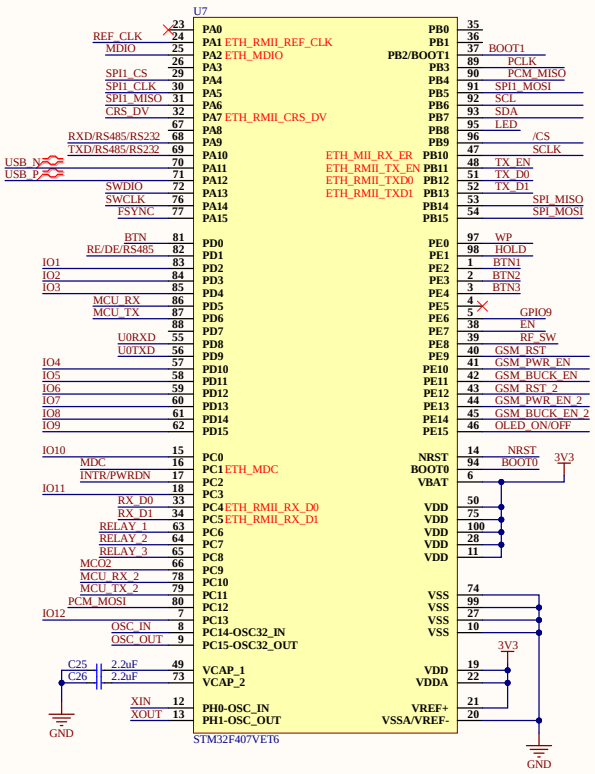


32.768KHZ CRYSTAL

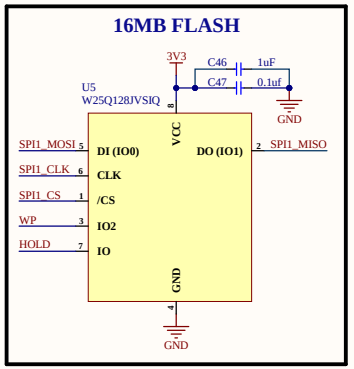


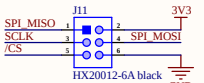


MCU

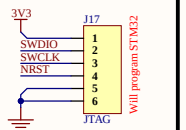


16MB FLASH

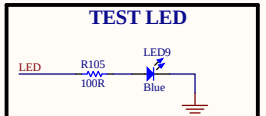




PROGRAMMING



TEST LED



ESP32 SHEET

GPIO9 — GPIO9
EN — EN
RF_SW — RF_SW
U0RXD — U0RXD
U0TXD — U0TXD

OLED SHEET

OLED_ON/OFF — OLED_ON/OFF
SCL — SCL
SDA — SDA
BTN1 — BTN1
BTN2 — BTN2
BTN3 — BTN3

RELAY_1 — RELAY_1
RELAY_2 — RELAY_2
RELAY_3 — RELAY_3

GSM_RST — GSM_RST
GSM_PWR_EN — GSM_PWR_EN
GSM_BUCK_EN — GSM_BUCK_EN

GSM_RST_2 — GSM_RST_2
GSM_PWR_EN_2 — GSM_PWR_EN_2
GSM_BUCK_EN_2 — GSM_BUCK_EN_2

3V3 — 3V3

GND — GND

IO_PORT: IO1 IO2 IO3 IO4 IO5 IO6 IO7 IO8 IO9 IO10 IO11 IO12 → IO_PORT

USB: USB_P, USB_N → USB

GSM_USB: USB_N, USB_P → GSM_USB

MCU_GSM: MCU_RX, MCU_TX, MCU_RX_2, MCU_TX_2 → MCU_GSM

MUX_RST/INT: RST_A, MUX1_INT → MUX_RST/INT

RS485/RS232: RXD/RS485/RS232, RE/DE/RS485, TXD/RS485/RS232 → RS485/RS232

PCM: PCM_MISO, PCM_MOSI, FSYNC, PCLK → PCM

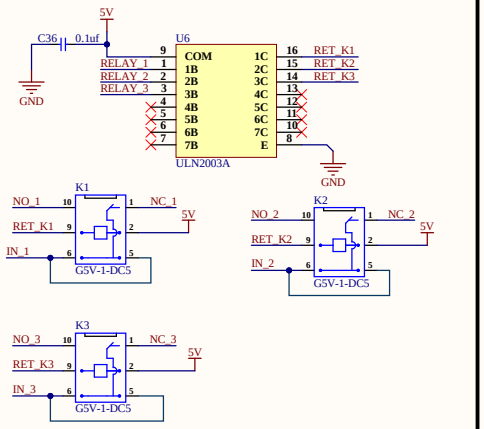
SPI: SPI_MISO, SPI_MOSI, SCLK, /CS → SPI

ETH_PHY: MDC, MDIO, CRS_DV, TX_D0, RX_D0, TX_D1, RX_D1, INTR/PWRDN, TX_EN, REF_CLK, MCO2 → ETH_PHY

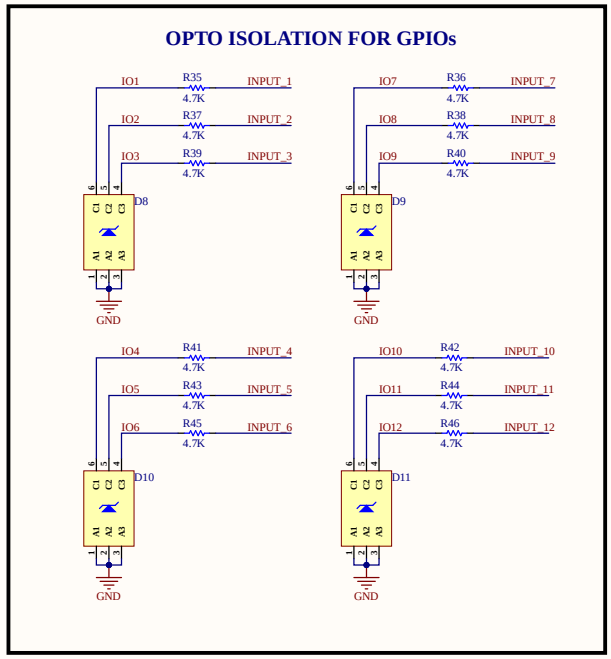
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Date: 3/19/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 12

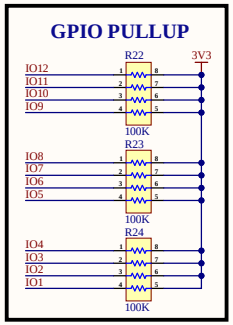
RELAY WITH DRIVER

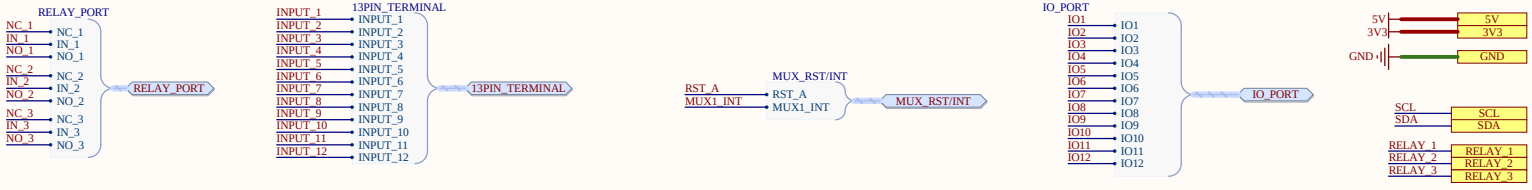


OPTO ISOLATION FOR GPIOs



GPIO PULLUP

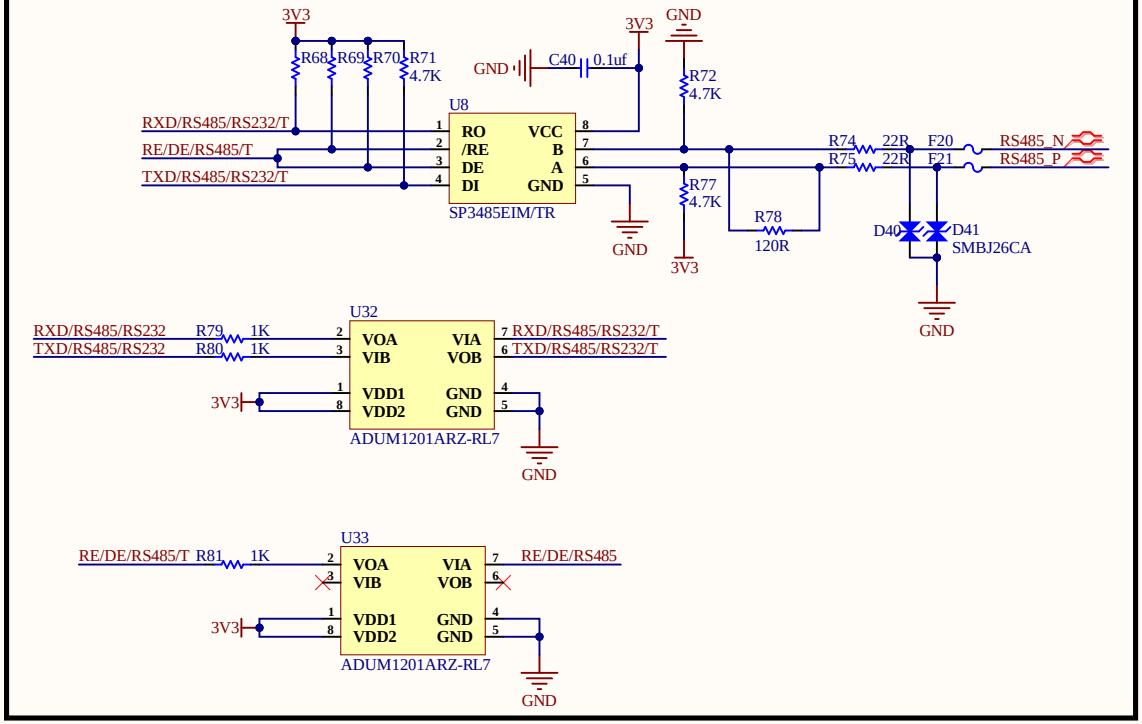




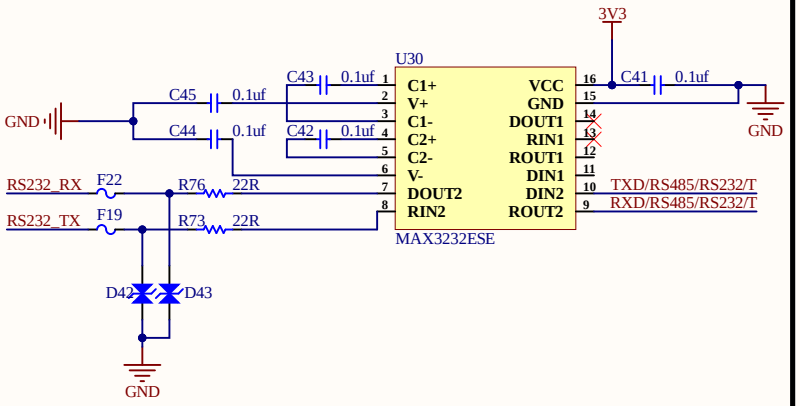
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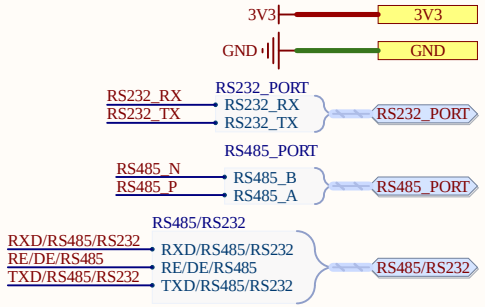
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Date: 3/14/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 12

RS485 to UART



RS232 to UART

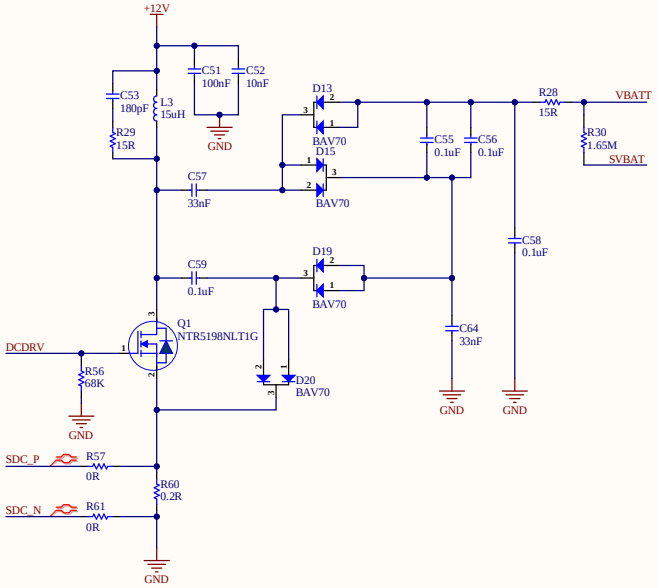




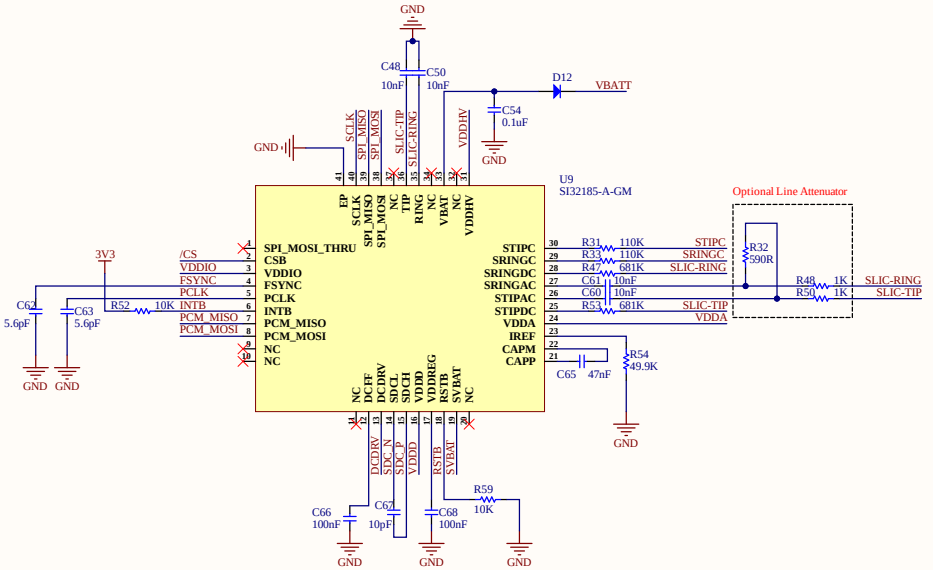
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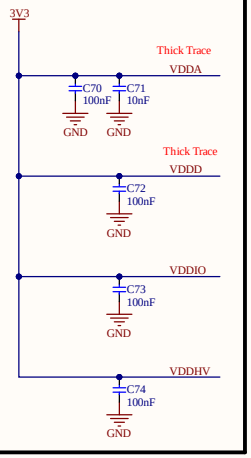
CAPACITIVE BOOST

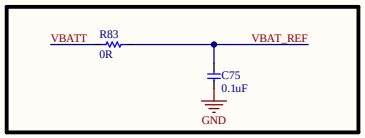


SI32185

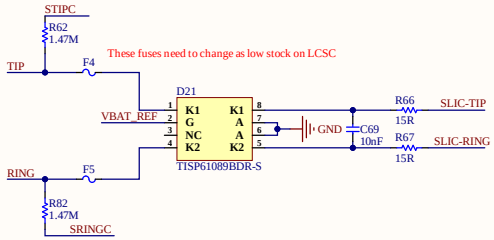


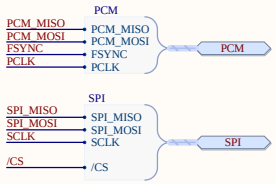
POWER

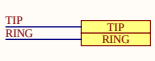




PROTECTION





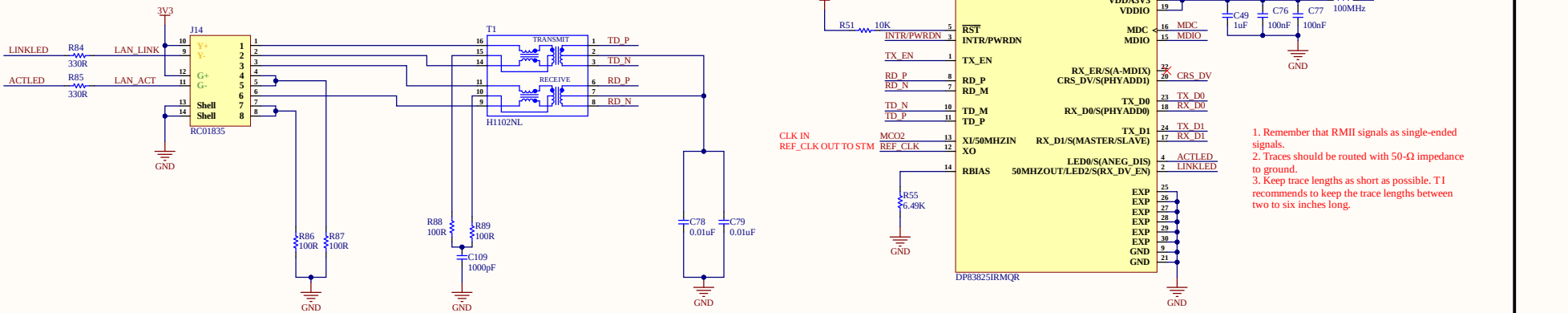


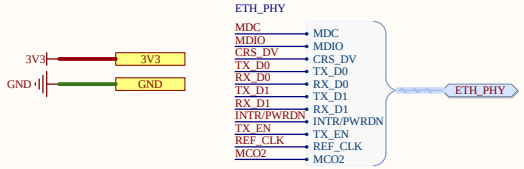


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Date: 3/19/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 12

ETHERNET/ DP83825IRMQR PHY

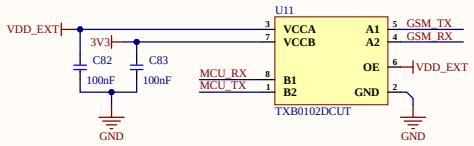




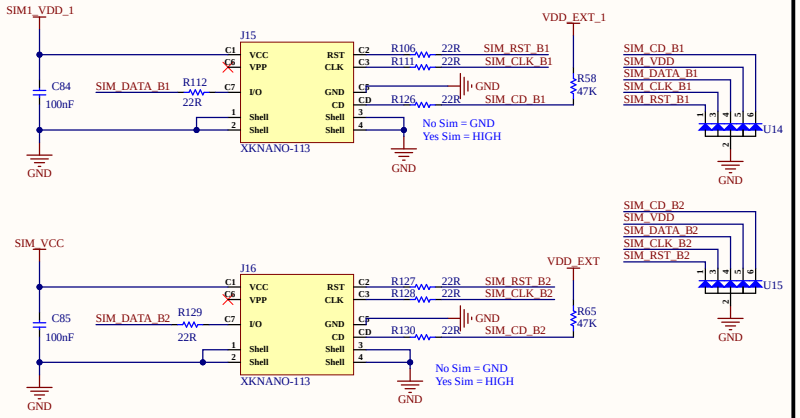
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Drawn By: Huzaifa Ali	Page Total: 12

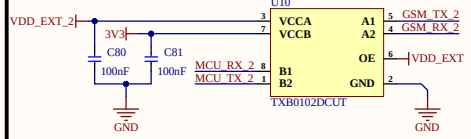
Level Shifter for GSM Serial 1



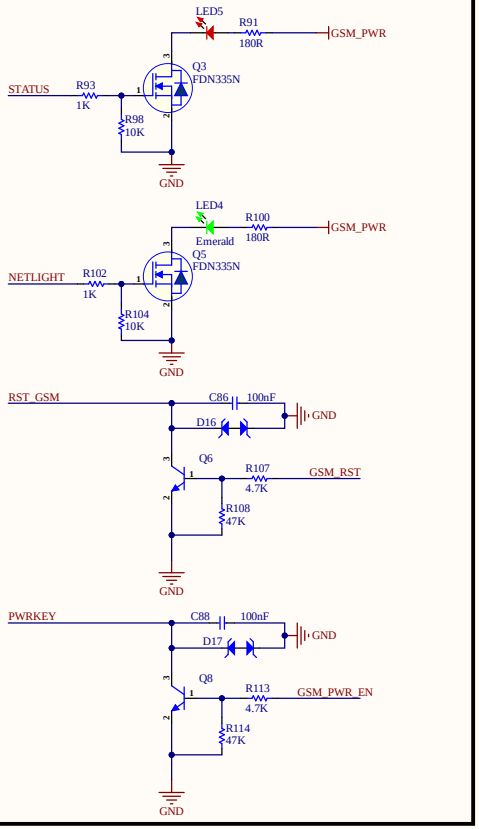
DUALSIM WITH MULTIPLEXER

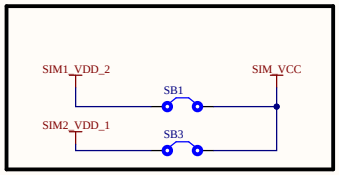


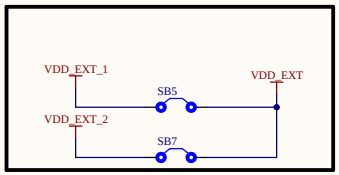
Level Shifter for GSM Serial 2



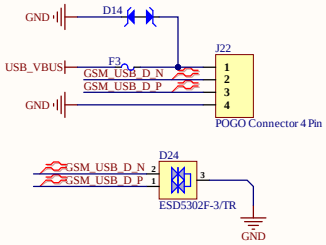
GSM_COMMANDS_1





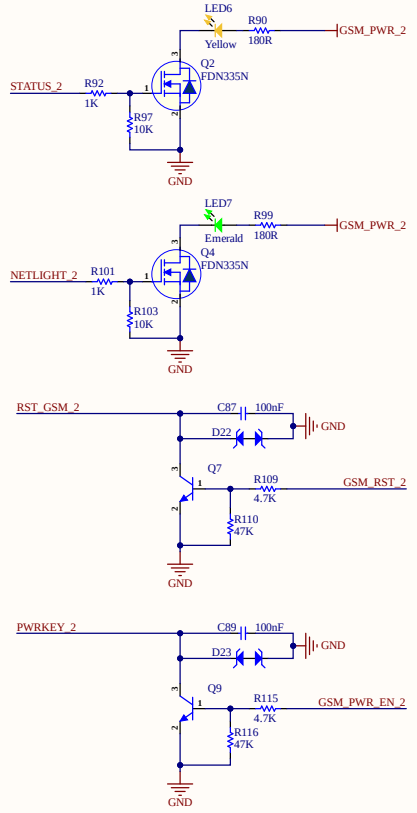


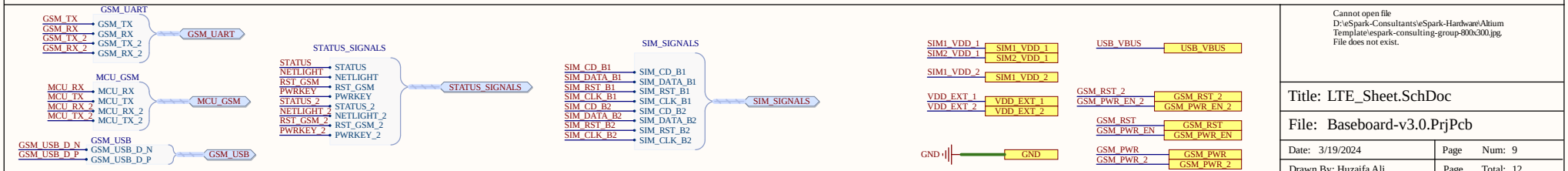
USB POGO for Updates



SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongl y recommended to follow these guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection
- Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

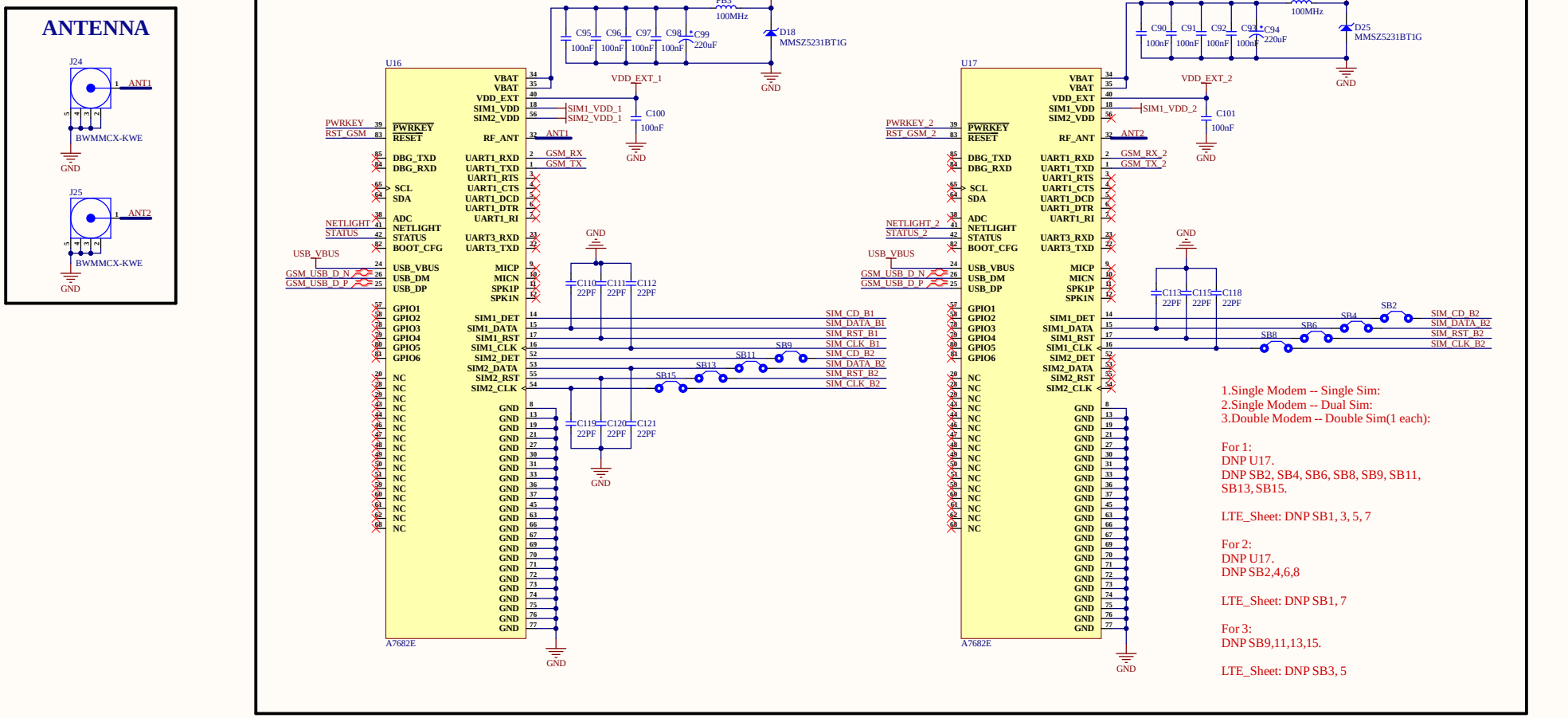
GSM_COMMANDS_2



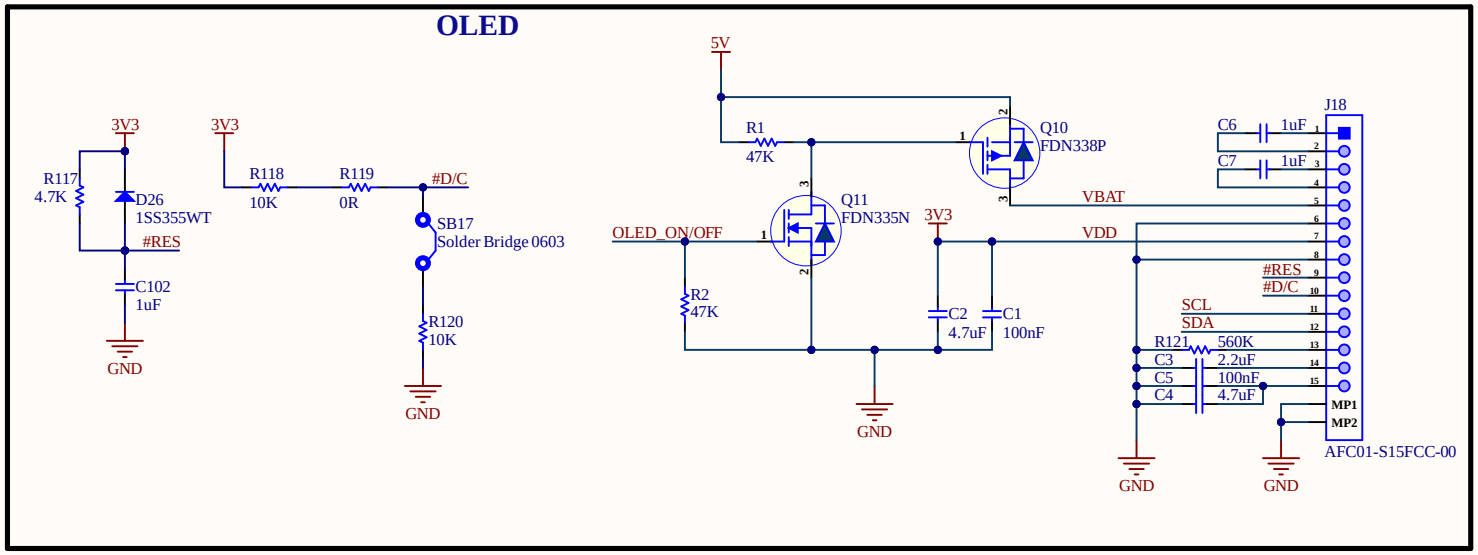


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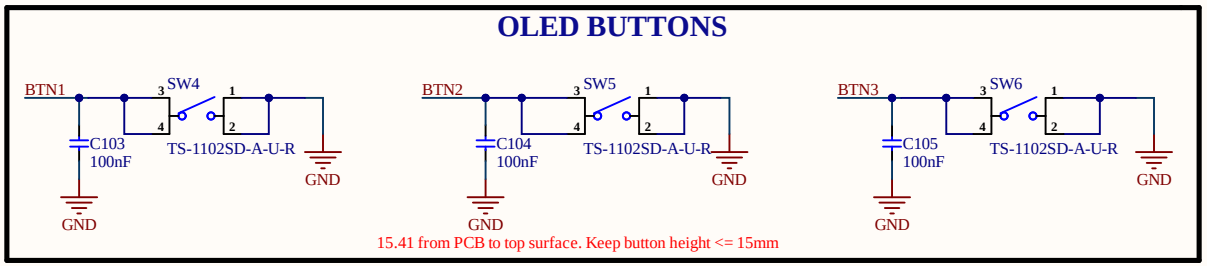
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ANTENNA
J24
ANT1
BWMMCX-KWE
GND
J25
ANT2
BWMMCX-KWE
GND
A7682E 01
A7682E 02
U16
U17
A7682E
GSM_PWR
GSM_PWR_2
FB3
FB4
100MHz
D18
D25
MMSZ5231BT1G
VDD_EXT_1
VDD_EXT_2
SIM1_VDD_1
SIM2_VDD_1
SIM1_VDD_2
C100
C101
100nF
ANT1
ANT2
GSM_RX
GSM_TX
GSM_RX_2
GSM_TX_2
PWRKEY
RST_GSM
PWRKEY_2
RST_GSM_2
NETLIGHT
STATUS
NETLIGHT_2
STATUS_2
SCL
SDA
BOOT_CFG
USB_VBUS
GSM_USB_D_N
GSM_USB_D_P
SIM_CD_B1
SIM_DATA_B1
SIM_RST_B1
SIM_CLK_B1
SIM_CD_B2
SIM_DATA_B2
SIM_RST_B2
SIM_CLK_B2
SB2
SB4
SB6
SB8
SB9
SB11
SB13
SB15
1.Single Modem -- Single Sim:
2.Single Modem -- Dual Sim:
3.Double Modem -- Double Sim(1 each):
For 1:
DNP U17.
DNP SB2, SB4, SB6, SB8, SB9, SB11, SB13, SB15.
LTE_Sheet: DNP SB1, 3, 5, 7
For 2:
DNP U17.
DNP SB2,4,6,8
LTE_Sheet: DNP SB1, 7
For 3:
DNP SB9,11,13,15.
LTE_Sheet: DNP SB3, 5
GSM_UART
STATUS_SIGNALS
GSM_USB
SIM_SIGNALS
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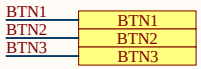
OLED

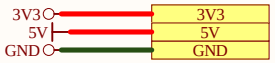


OLED BUTTONS





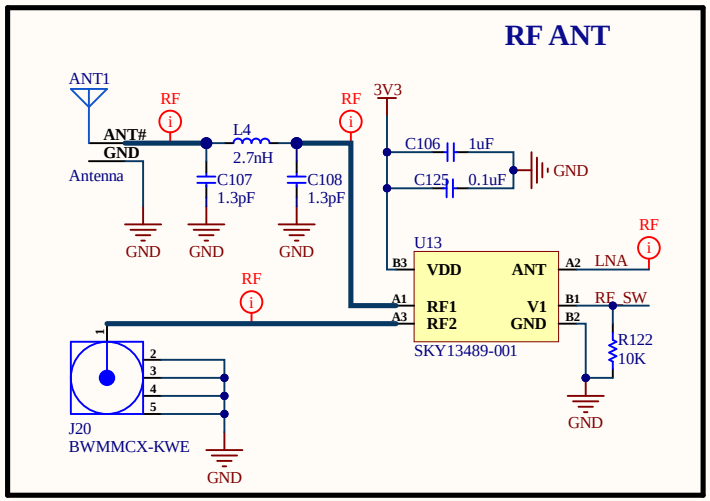


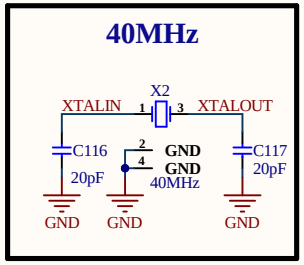


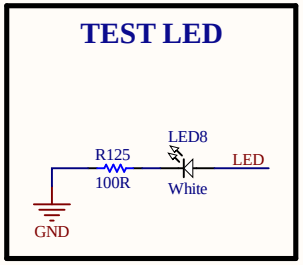
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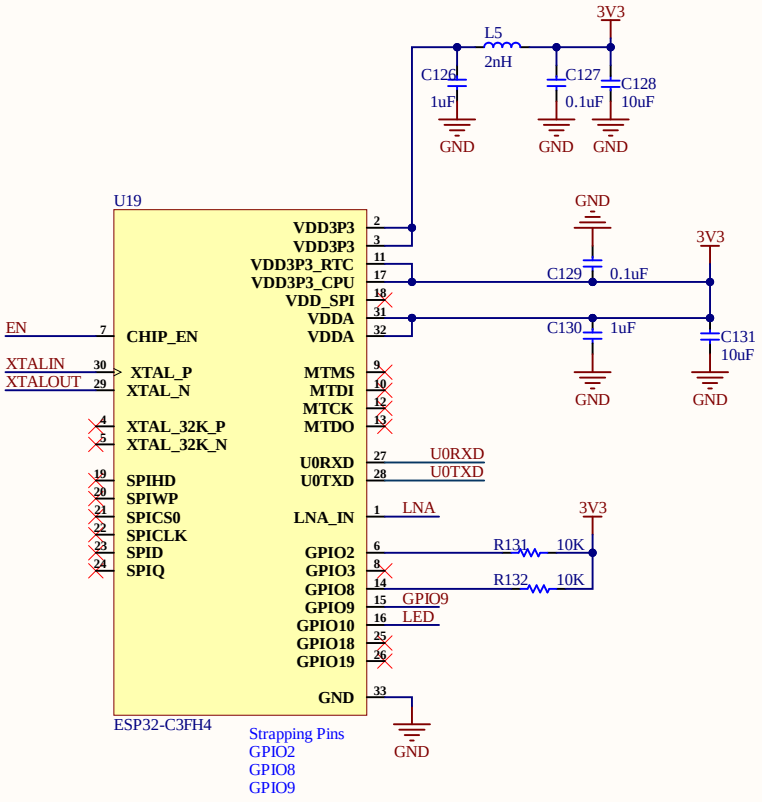
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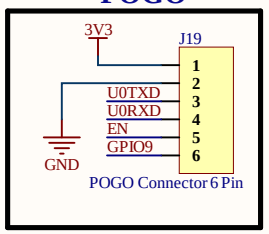
ESP32-C3FH4

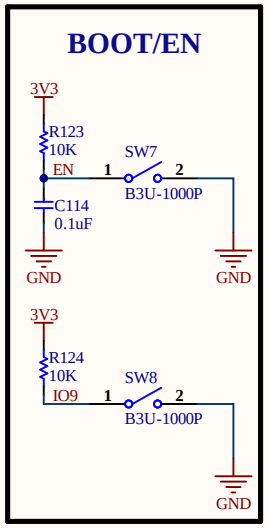


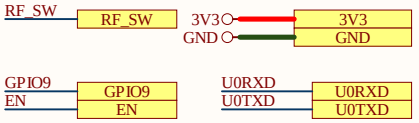












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